

RESEARCH

& INNOVATION UPDATES

AI motion control enables robots to adapt to diverse inputs



SONIC enables diverse humanoid tasks through a universal control policy that handles diverse input modalities and control interfaces (Credit: <https://nvlabs.github.io/SONIC>)

Robots gaining human-like motion could soon automate dynamic tasks from logistics to physical assistance, thanks to Nvidia's SONIC AI motion control system that unifies whole-body movement across behaviours such as running, crawling, and balance responses. SONIC's scalable model reduces development time and enables robots to adapt to diverse inputs, such as VR control or high-level cues, boosting utility in teleoperation, industrial automation, and service robots beyond the lab.

Circuits on shrinkable films produce retrofit smart tech

A low-cost way to print liquid-metal circuits on shrinkable films, enables compact 3D electronics that contour to complex shapes for adaptive antennas, wearable health sensors, and smart home interfaces. Penn State (Pennsylvania) researchers in the US have demonstrated heat-induced shrinking turns flat circuits into durable, conformal devices, shown in a gesture-sensing wearable ring prototype, pointing to scalable, retrofit smart tech across biomedical and automation uses.

As a proof-of-concept test device, the team developed a wearable ring with an embedded, miniaturised accelerometer that captured and successfully communicated gesture-based movements over a network (Credit: Courtney Robinson/Penn State)

